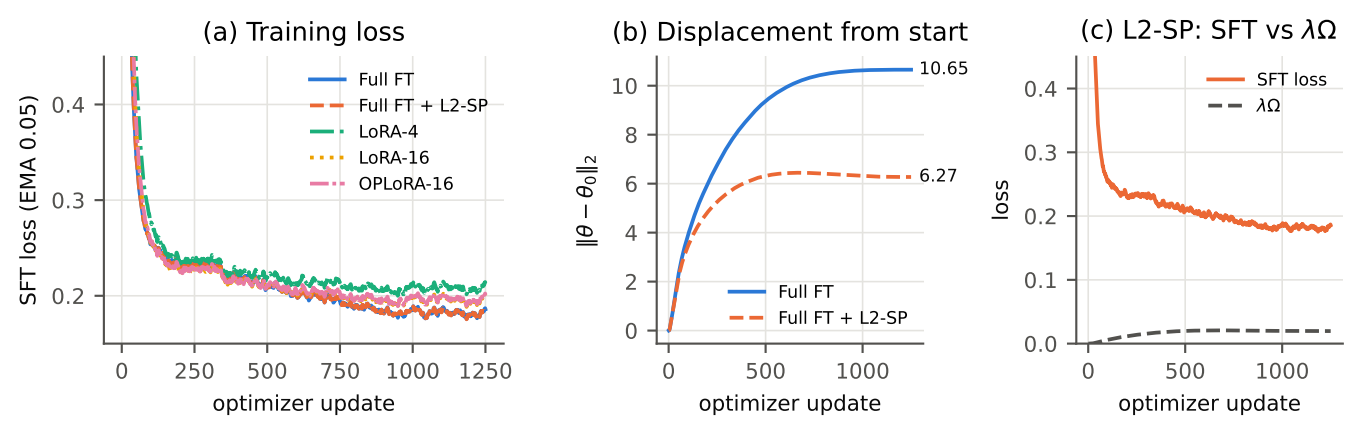

(a) Training loss
SFT loss (EMA 0.05)
0.4
0.3
0.2
0
250
500
750
1000
1250
optimizer update
Full FT
Full FT + L2-SP
LoRA-4
LoRA-16
OPLoRA-16
(b) Displacement from start
$\|\theta - \theta_0\|_2$
10
8
6
4
2
0
0
500
1000
optimizer update
10.65
6.27
Full FT
Full FT + L2-SP
(c) L2-SP: SFT vs $\lambda\Omega$
loss
0.4
0.3
0.2
0.1
0.0
0
500
1000
optimizer update
SFT loss
$\lambda\Omega$